Supplementary Appendix

This appendix has been provided by the authors to give readers additional information about their work.

Supplement to: Socinski MA, Jotte RM, Cappuzzo F, et al. Atezolizumab for first-line treatment of metastatic nonsquamous NSCLC. N Engl J Med 2018;378:2288-301. DOI: 10.1056/NEJMoa1716948

SUPPLEMENTARY APPENDIX

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Supplement to: Socinski MA, Jotte RM, Cappuzzo F, et al. Atezolizumab in First-Line Metastatic Non-Squamous Non-Small-Cell Lung Cancer

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Supplementary Methods

The Teff gene signature was defined by *PD-L1 (CD274)*, *CXCL9*, and *IFN-γ* mRNA expression using RNA isolated from baseline, macro-dissected tumor tissue and measured with a Clinical Trial Assay based on real-time qPCR (Roche Molecular Systems). The Teff assay was transferred and qualified at a central laboratory (Targos) prior to retrospective IMpower150 patient sample testing. Patients were included in the Teff-high population if the Teff gene signature score was ≥–1.91 or in the Teff-low population if the Teff signature score was <–1.91. Additional Teff cut-offs (–2.38, –2.93) were also evaluated as part of the secondary endpoint analysis. The Teff cut-offs were pre-specified in IMpower150 study SAP.

Figure S1. Alpha-Spending Algorithm. The study design and α-spending algorithm to control for type I error for PFS and OS in the primary analysis populations for the statistical testing of ABCP vs. BCP followed by ACP vs. BCP is depicted. ABCP denotes atezolizumab + bevacizumab + carboplatin + paclitaxel; BCP, bevacizumab + carboplatin + paclitaxel; ITT, intention-to-treat; OS, overall survival; PFS, progression-free survival; Teff, T-effector gene-signature expression; WT, wild type.

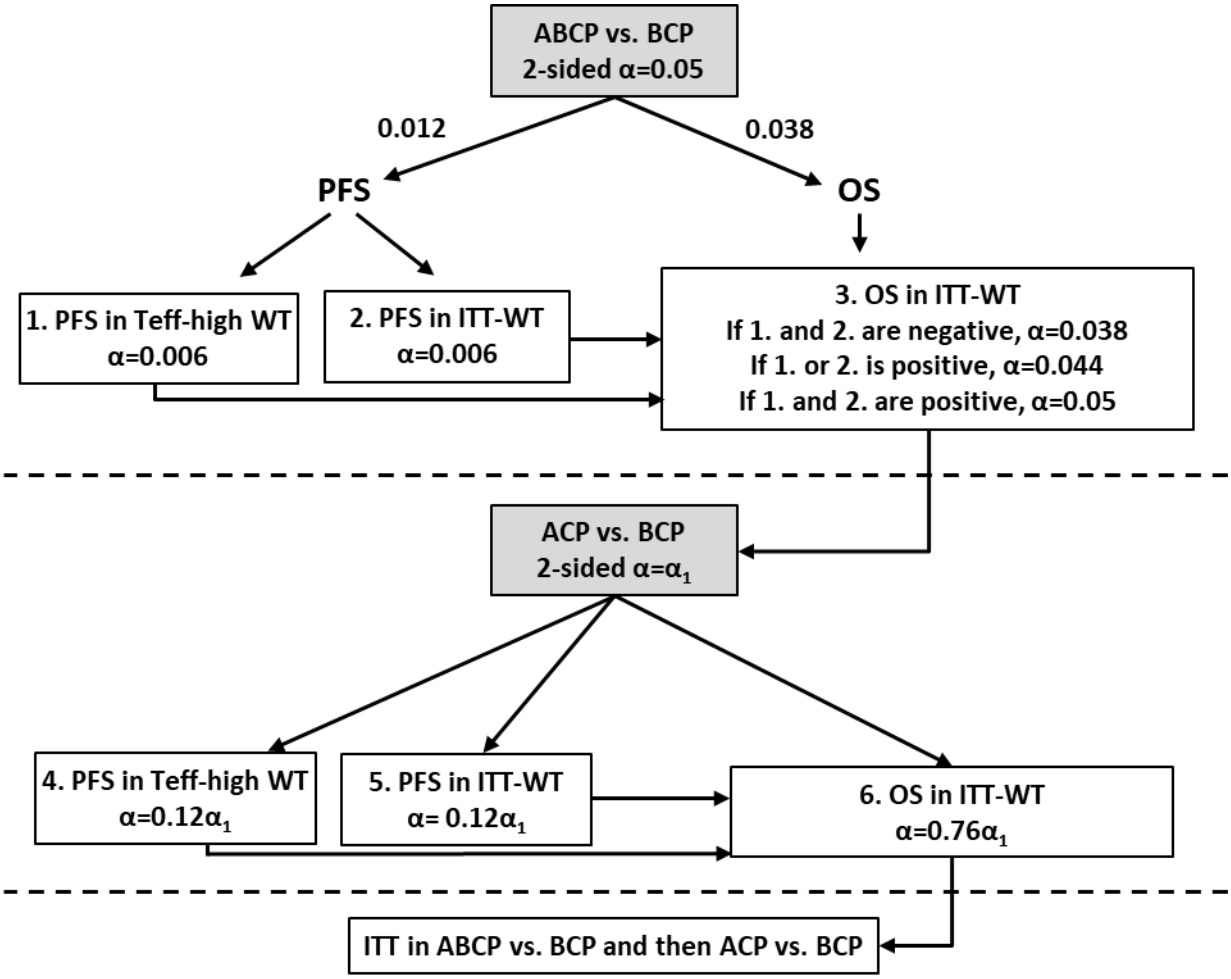
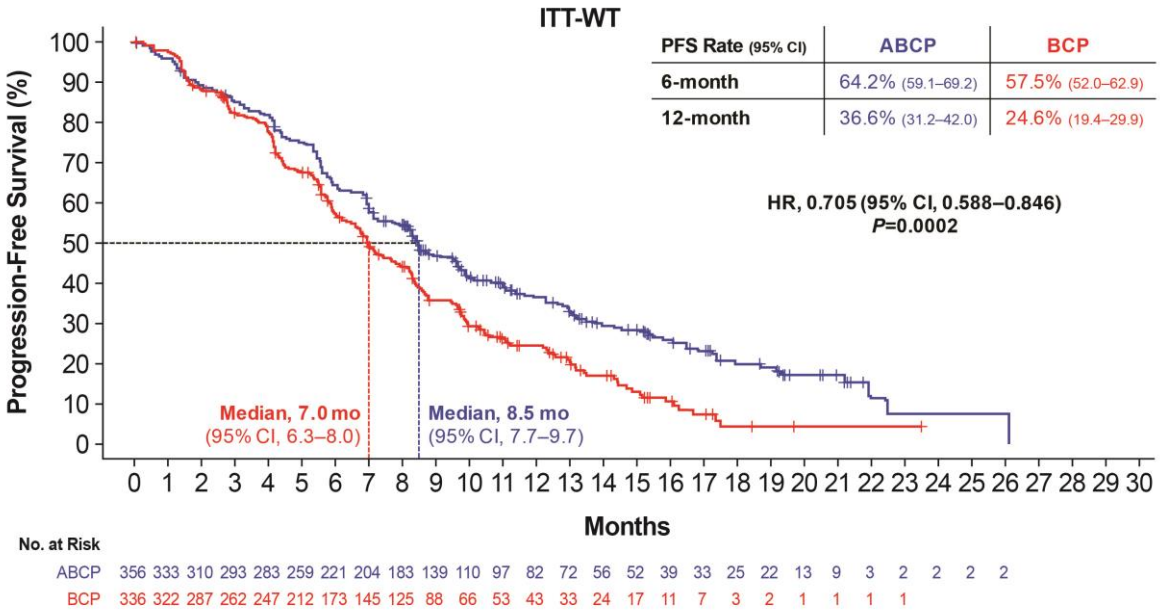


Figure S2. Independent Review Facility-Assessed Progression-Free Survival Between Atezolizumab, Bevacizumab, Carboplatin, and Paclitaxel and Bevacizumab, Carboplatin, and Paclitaxel in the ITT-WT and Teff-high WT Populations.

Panel A shows Kaplan-Meier estimates of progression-free survival in the ITT-WT population. Panel B shows Kaplan-Meier estimates of progression-free survival in the ITT-Teff-high WT population. Stratified HRs are given for the ITT-WT and Teff-high WT. ABCP denotes atezolizumab + bevacizumab + carboplatin + paclitaxel; BCP, bevacizumab + carboplatin + paclitaxel; HR, hazard ratio; ITT, intention-to-treat; PFS, progression-free survival; Teff, T-effector gene-signature expression; WT, wild type. Data cutoff: 15 September 2017.

A)



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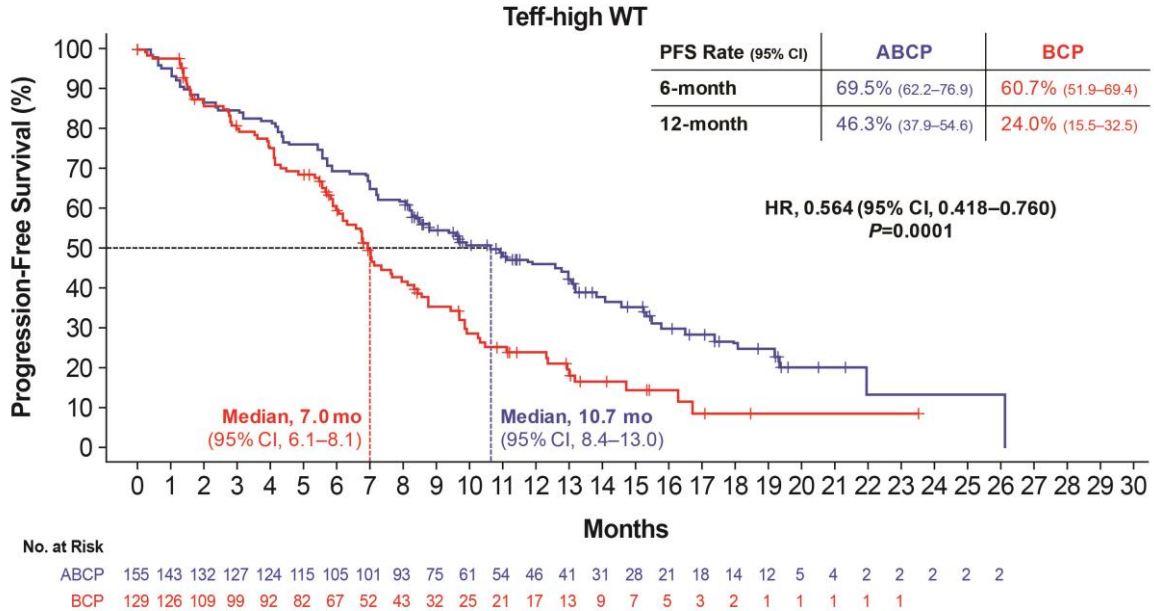
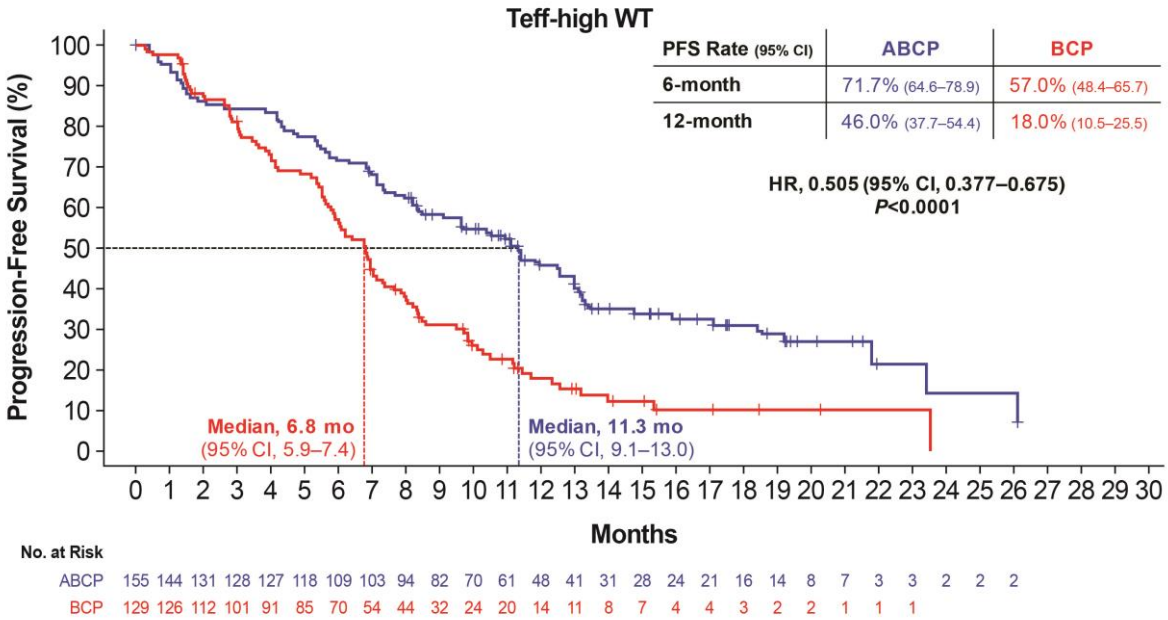


Figure S3. Investigator-Assessed Progression-Free Survival Between Atezolizumab, Bevacizumab, Carboplatin, and Paclitaxel and Bevacizumab, Carboplatin, and Paclitaxel in the Teff-high WT and Teff-low WT Populations.

Panel A shows Kaplan-Meier estimates of progression-free survival in the Teff-high WT population. Panel B shows Kaplan-Meier estimates of progression-free survival in the Teff-low WT population. Stratified HR for Teff-high WT; unstratified HR for Teff-low WT. ABCP denotes atezolizumab + bevacizumab + carboplatin + paclitaxel; BCP, bevacizumab + carboplatin + paclitaxel; HR, hazard ratio; PFS, progression-free survival; WT, wild type. Data cutoff: 15 September 2017.

A)



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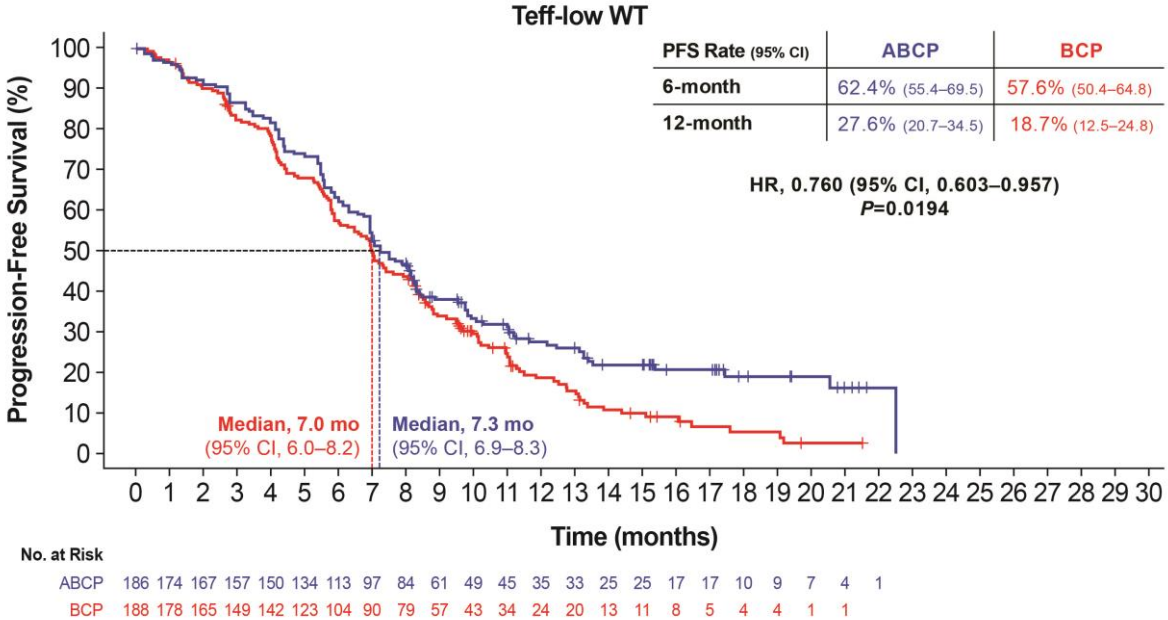
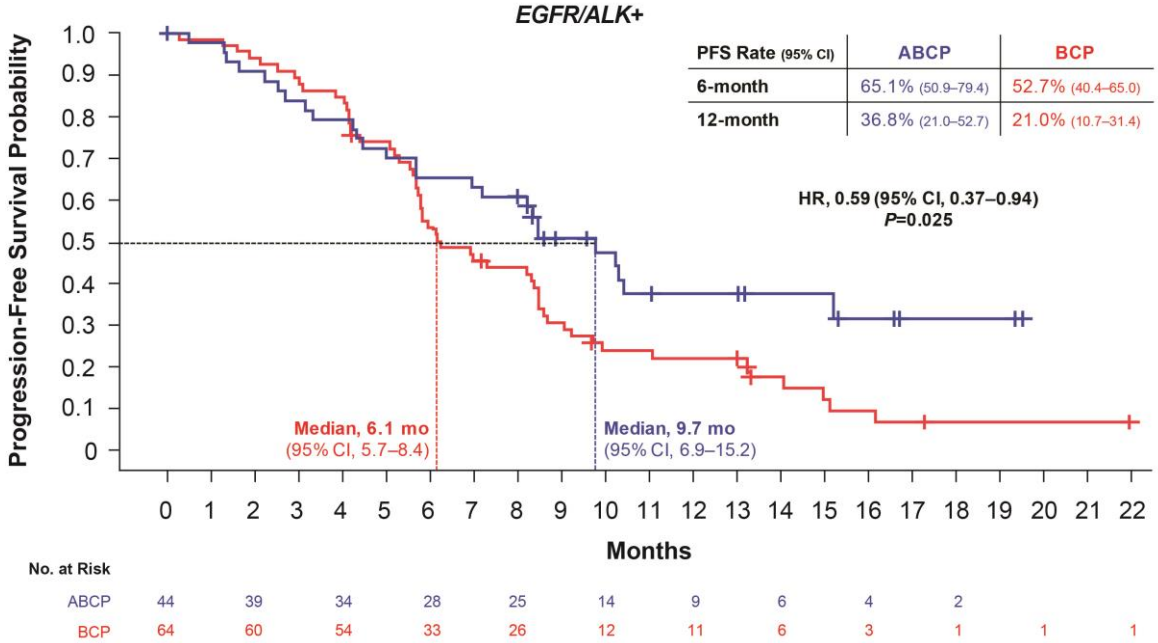


Figure S4. Investigator-Assessed Progression-Free Survival Between Atezolizumab, Bevacizumab, Carboplatin, and Paclitaxel and Bevacizumab, Carboplatin, and Paclitaxel in the ITT.

Panel A shows Kaplan-Meier estimates of progression-free survival in patients with *EGFR* or *ALK* genomic alterations. Panel B shows Kaplan-Meier estimates of progression-free survival in the ITT population, which includes patients with *EGFR* mutation or *ALK* translocation. Stratified HR for ITT; unstratified HR for patients with *EGFR* or *ALK* genomic alterations. ABCP denotes atezolizumab + bevacizumab + carboplatin + paclitaxel; BCP, bevacizumab + carboplatin + paclitaxel; HR, hazard ratio; ITT, intention-to-treat; PFS, progression-free survival. Data cutoff: 15 September 2017.

A)



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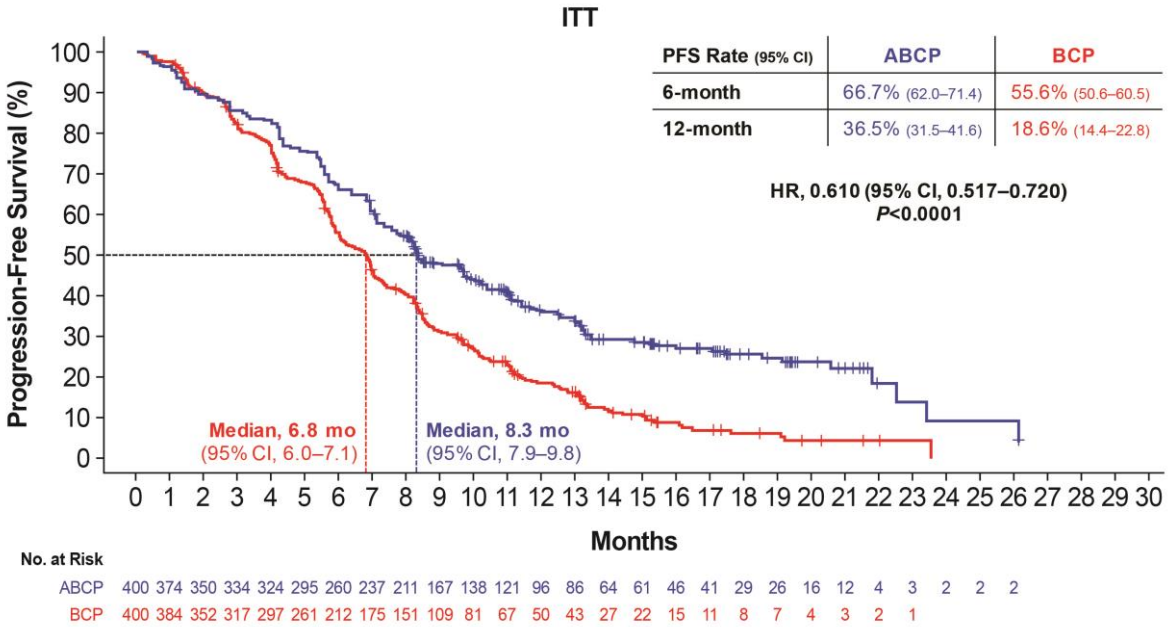
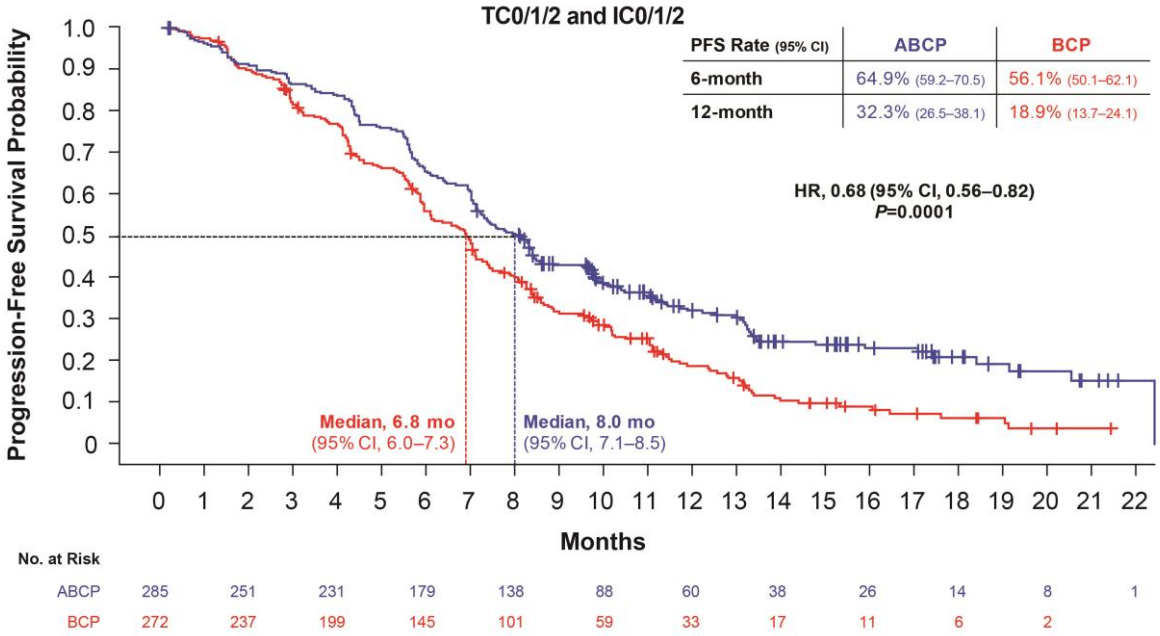


Figure S5. Investigator-Assessed Progression-Free Survival Between Atezolizumab, Bevacizumab, Carboplatin, and Paclitaxel and Bevacizumab, Carboplatin, and Paclitaxel in the PD-L1–Low/Negative and PD-L1–High IHC Subgroups. Panel A shows Kaplan-Meier estimates of progression-free survival in the PD-L1–low/negative (TC0/1/2 and IC0/1/2) subgroup of the ITT-WT population. Panel B shows Kaplan-Meier estimates of progression-free survival in the PD-L1–high (TC3 or IC3) subgroup of the ITT-WT population. Unstratified HRs. TC0/1/2 and IC0/1/2 = PD-L1 expression on < 50% of TC and < 10% of IC; TC3 or IC3 = PD-L1 expression on ≥ 50% of TC or ≥ 10% of IC. ABCP denotes atezolizumab + bevacizumab + carboplatin + paclitaxel; BCP, bevacizumab + carboplatin + paclitaxel; HR, hazard ratio; IC, tumor-infiltrating immune cells; IHC, immunohistochemistry; ITT, intention-to-treat; PD-L1, programmed death-ligand 1; PFS, progression-free survival; TC, tumor cells; WT, wild type. Data cutoff: 15 September 2017.

A)



B)

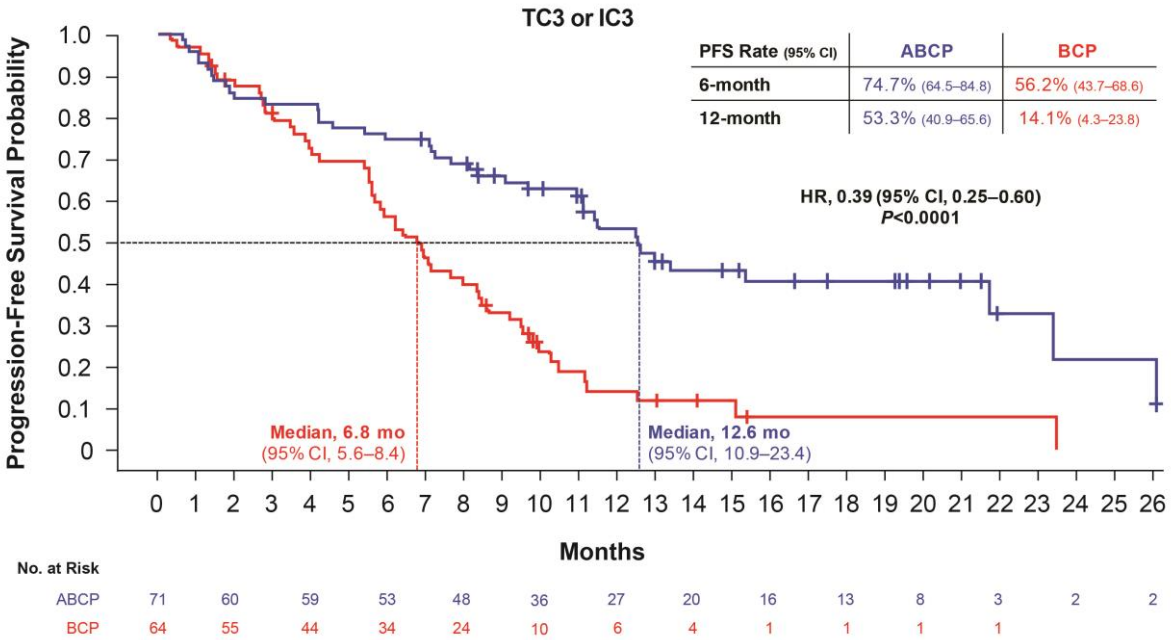
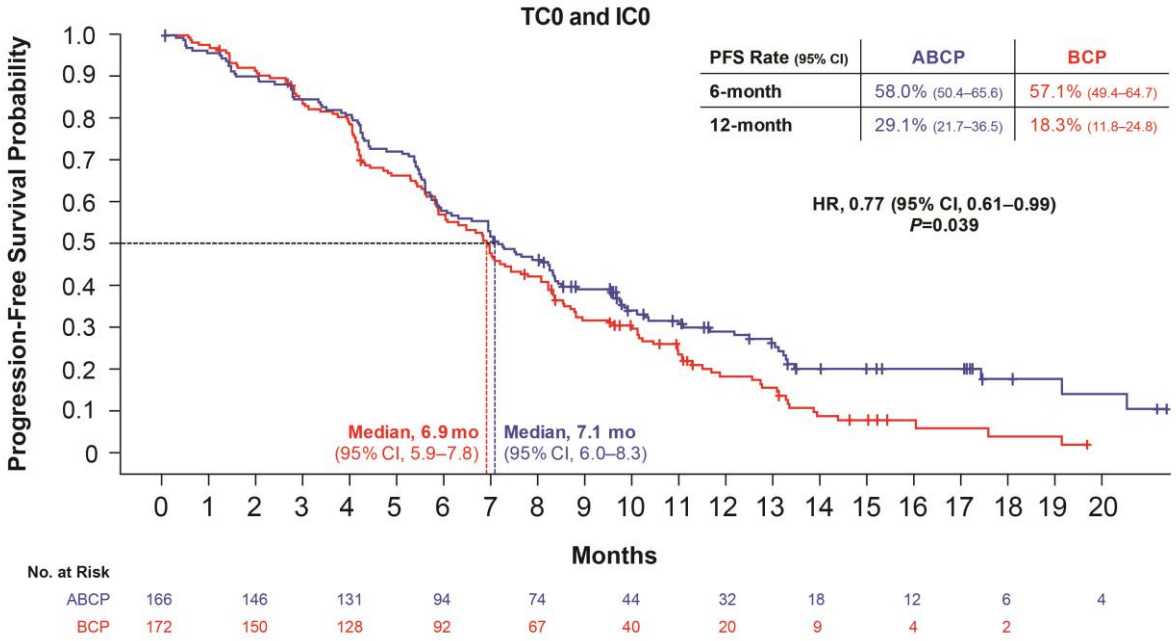

TC3 or IC3
Progression-Free Survival Probability
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
PFS Rate (95% CI)
ABCP
BCP
6-month
74.7% (64.5–84.8)
56.2% (43.7–68.6)
12-month
53.3% (40.9–65.6)
14.1% (4.3–23.8)
HR, 0.39 (95% CI, 0.25–0.60)
P<0.0001
Median, 6.8 mo
(95% CI, 5.6–8.4)
Median, 12.6 mo
(95% CI, 10.9–23.4)
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26
Months
No. at Risk
ABCP 71 60 59 53 48 36 27 20 16 13 8 3 2 2
BCP 64 55 44 34 24 10 6 4 1 1 1 1

Figure S6. Investigator-Assessed Progression-Free Survival Between Atezolizumab, Bevacizumab, Carboplatin, and Paclitaxel and Bevacizumab, Carboplatin, and Paclitaxel in the PD-L1–Negative and PD-L1–Low IHC Subgroups.

Panel A shows Kaplan-Meier estimates of progression-free survival in the PD-L1–negative (TC0 and IC0) IHC subgroup in the ITT-WT population. Panel B shows Kaplan-Meier estimates of progression-free survival in the PD-L1–low (TC1/2 or IC1/2) IHC subgroup excluding the PD-L1–high (TC3 or IC3) subgroup in the ITT-WT population. Unstratified HRs. TC1/2 or IC1/2 = PD-L1 expression on ≥ 1% and < 50% of TC or < 10% of IC; TC0 and IC0 = PD-L1 expression on < 1% of TC and IC. ABCP denotes atezolizumab + bevacizumab + carboplatin + paclitaxel; BCP, bevacizumab + carboplatin + paclitaxel; HR, hazard ratio; IC, tumor-infiltrating immune cells; IHC, immunohistochemistry; ITT, intention-to-treat; PD-L1, programmed death-ligand 1; PFS, progression-free survival; TC, tumor cells; WT, wild type. Data cutoff: 15 September 2017.

A)



B)

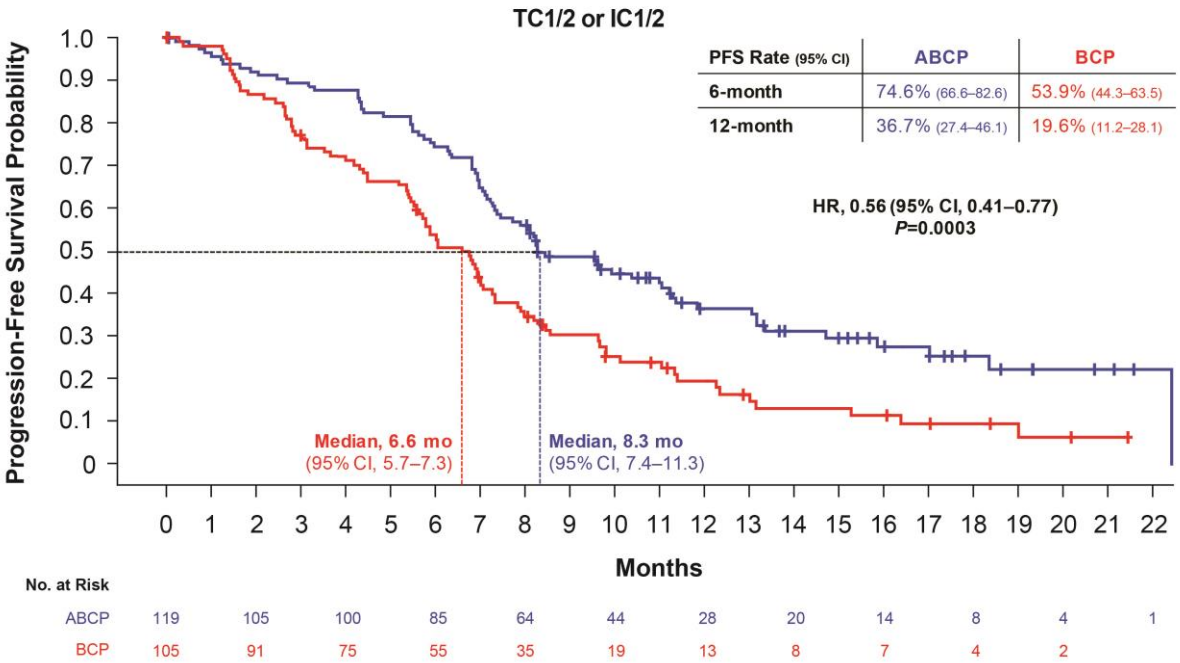

TC1/2 or IC1/2
Progression-Free Survival Probability
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
PFS Rate (95% CI)
ABCP
BCP
6-month
74.6% (66.6–82.6)
53.9% (44.3–63.5)
12-month
36.7% (27.4–46.1)
19.6% (11.2–28.1)
HR, 0.56 (95% CI, 0.41–0.77)
P=0.0003
Median, 6.6 mo
(95% CI, 5.7–7.3)
Median, 8.3 mo
(95% CI, 7.4–11.3)
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22
Months
No. at Risk
ABCP 119 105 100 85 64 44 28 20 14 8 4 1
BCP 105 91 75 55 35 19 13 8 7 4 2

Figure S7. Investigator-Assessed Progression-Free Survival Between Atezolizumab, Bevacizumab, Carboplatin, and Paclitaxel and Bevacizumab, Carboplatin, and Paclitaxel in Clinical Characteristic and Biomarker Subgroups.

Hazard ratios and 95% confidence intervals for investigator-assessed progression-free survival in clinical characteristic and biomarker subgroups of the ITT-WT population. Stratified HR for ITT-WT; unstratified HR for all other subgroups. ABCP denotes atezolizumab + bevacizumab + carboplatin + paclitaxel; BCP, bevacizumab + carboplatin + paclitaxel; ECOG, Eastern Cooperative Oncology Group; HR, hazard ratio; ITT, intention-to-treat; PFS, progression-free survival; WT, wild type. Data cutoff: 15 September 2017.

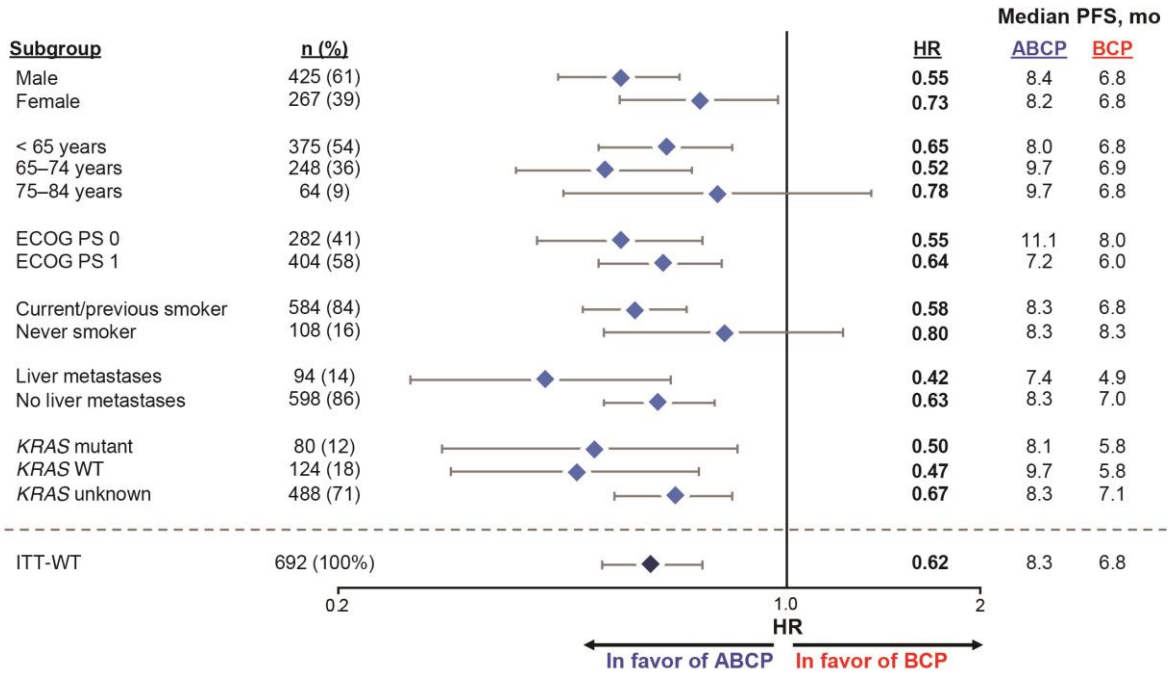


Table S1. PD-L1 Scoring of Tumor Cells and Tumor-Infiltrating Immune Cells.

Score	Percentage of PD-L1–Expressing Cells
TC3 or IC3	≥ 50% of TC or ≥ 10% of IC
TC2/3 or IC2/3	≥ 5% of TC or IC
TC1/2/3 or IC1/2/3	≥ 1% of TC or IC
TC1/2 or IC1/2	≥ 1% of TC or IC and < 50% of TC or < 10% of IC
TC0/1/2 and IC0/1/2	< 50% of TC and < 10% of IC
TC0 and IC0	< 1% of TC and IC

IC denotes tumor-infiltrating immune cells; PD-L1, programmed death-ligand 1; TC, tumor cells.

Table S2. Teff Gene Signature Expression and PD-L1 Expression Subgroup Prevalence at Baseline in the ITT Population.

	ITT	
Characteristic	**ABCP (N=400)**	**BCP (N=400)**
Teff gene signature expression — no. (%)*		
High	166 (41.5)	148 (37.0)
Low	217 (54.3)	231 (57.8)
Unknown	17 (4.3)	21 (5.3)
PD-L1 subgroups — no. (%)		
TC3 or IC3	75 (18.8)	73 (18.3)
TC2/3 or IC2/3	140 (35.0)	133 (33.3)
TC1/2/3 or IC1/2/3	209 (52.3)	195 (48.8)
TC1/2 or IC1/2	135 (33.8)	127 (31.8)
TC0/1/2 and IC0/1/2	325 (81.3)	327 (81.8)
TC0 and IC0	190 (47.5)	200 (50.0)

* The Teff gene signature cutoff of –1.91 was used.
TC3 or IC3 = PD-L1 expression on ≥ 50% of TC or ≥ 10% of IC; TC2/3 or IC2/3 = PD-L1 expression on ≥ 5% of TC or IC; TC1/2/3 or IC1/2/3 = PD-L1 expression on ≥ 1% of TC or IC; TC1/2 or IC1/2 = PD-L1 expression on ≥ 1% of TC or IC and < 50% of TC or < 10% of IC; TC0/1/2 and IC0/1/2 = PD-L1 expression on < 50% of TC and < 10% of IC; TC0 and IC0 = PD-L1 expression on < 1% of TC and IC.
ABCP denotes atezolizumab + bevacizumab + carboplatin + paclitaxel; BCP, bevacizumab + carboplatin + paclitaxel; IC, tumor-infiltrating immune cell; ITT, intention-to-treat; PD-L1, programmed death-ligand 1; TC, tumor cell; Teff, T-effector gene-signature expression.
Data cutoff: 15 September 2017.

Table S3. Baseline Characteristics of the Primary Analysis Populations.

	ITT-WT		Teff-high WT	
Characteristic	**ABCP (n=356)**	**BCP (n=336)**	**ABCP (n=155)**	**BCP (n=129)**
Age — yr				
Median	63	63	66	63
Range	31–89	41–87	31–89	41–82
Age groups — no. (%)				
<65 yr	191 (53.7)	184 (54.8)	71 (45.8)	74 (57.4)
65 to 74 yr	130 (36.5)	118 (35.1)	69 (44.5)	44 (34.1)
75 to 84 yr	32 (9.0)	32 (9.5)	12 (7.7)	11 (8.5)
≥85 yr	3 (0.8)	2 (0.6)	3 (1.9)	0
Sex — no. (%)				
Male	217 (61.0)	208 (61.9)	94 (60.6)	74 (57.4)
Liver metastases at enrollment — no. (%)				
No	309 (86.8)	289 (86.0)	137 (88.4)	115 (89.1)
Teff gene-signature expression — no. (%)*				
High	155 (43.5)	129 (38.4)	155 (100)	129 (100)
Low	186 (52.2)	188 (56.0)	0	0
Unknown	15 (4.2)	19 (5.7)	0	0
PD-L1 subgroups — no. (%)				
TC3 or IC3	71 (19.9)	64 (19.0)	63 (40.6)	47 (36.4)
TC2/3 or IC2/3	129 (36.2)	115 (34.2)	100 (64.5)	79 (61.2)
TC1/2/3 or IC1/2/3	190 (53.4)	164 (48.8)	124 (80.0)	101 (78.3)
TC1/2 or IC1/2	119 (33.4)	105 (31.3)	61 (39.4)	56 (43.4)
TC0/1/2 and IC0/1/2	285 (80.1)	272 (81.0)	92 (59.4)	82 (63.6)
TC0 and IC0	166 (46.6)	167 (49.7)	31 (20.0)	28 (21.7)
Race — no. (%)				
White	294 (82.6)	296 (88.1)	125 (80.6)	106 (82.2)
Asian	42 (11.8)	23 (6.8)	22 (14.2)	13 (10.1)
Black	3 (0.8)	11 (3.3)	2 (1.3)	6 (4.7)
American Indian or Alaska Native	3 (0.8)	1 (0.3)	0	1 (0.8)
Multiple	2 (0.6)	0	1 (0.6)	0
Unknown	12 (3.4)	5 (1.5)	5 (3.2)	3 (2.3)
ECOG performance status — no. (%)				
0	139 (39.0)	143 (42.6)	57 (36.8)	64 (49.6)

1	214 (60.1)	190 (56.5)	98 (63.2)	65 (50.4)
Other than 0/1	3 (0.8)	3 (0.9)	0	0
Tobacco use history — no. (%)				
Never	58 (16.3)	50 (14.9)	22 (14.2)	19 (14.7)
Current	83 (23.3)	84 (25.0)	29 (18.7)	29 (22.5)
Previous	215 (60.4)	202 (60.1)	104 (67.1)	81 (62.8)
Non-squamous histology — no. (%)				
Adenocarcinoma	335 (94.1)	315 (93.8)	145 (93.5)	118 (91.5)
Other†	18 (5.1)	15 (4.5)	10 (6.5)	7 (5.4)
Unknown or not assessed	3 (0.8)	6 (1.8)	0	4 (3.1)
EGFR mutation status — no. (%)				
Negative	344 (96.6)	328 (97.6)	150 (96.8)	128 (99.2)
Unknown	12 (3.4)	8 (2.4)	5 (3.2)	1 (0.8)
EML4-ALK rearrangement status — no. (%)				
Negative	353 (99.2)	334 (99.4)	154 (99.4)	129 (100)
Unknown	3 (0.8)	2 (0.6)	1 (0.6)	0
KRAS mutation status — no. (%)				
Positive	44 (12.4)	36 (10.7)	20 (12.9)	15 (11.6)
Negative	55 (15.4)	69 (20.5)	21 (13.5)	27 (20.9)
Unknown	257 (72.2)	231 (68.8)	114 (73.5)	87 (67.4)

* The Teff gene signature cut-off of –1.91 was used.
† Other includes adenocarcinoma with neuroendocrine features, adenosquamous, bronchioloalveolar carcinoma, large cell, sarcomatoid, and undifferentiated.
TC3 or IC3 = PD-L1 expression on ≥ 50% of TC or ≥ 10% of IC; TC2/3 or IC2/3 = PD-L1 expression on ≥ 5% of TC or IC; TC1/2/3 or IC1/2/3 = PD-L1 expression on ≥ 1% of TC or IC; TC1/2 or IC1/2 = PD-L1 expression on ≥ 1% of TC or IC and < 50% of TC or < 10% of IC; TC0/1/2 and IC0/1/2 = PD-L1 expression on < 50% of TC and < 10% of IC; TC0 and IC0 = PD-L1 expression on < 1% of TC and IC.
ABCP denotes atezolizumab + bevacizumab + carboplatin + paclitaxel; BCP, bevacizumab + carboplatin + paclitaxel; ECOG, Eastern Cooperative Oncology Group; IC, tumor-infiltrating immune cell; ITT, intention-to-treat; PD-L1, programmed death-ligand 1; TC, tumor cell; Teff, T-effector gene-signature expression; WT, wild type.
Data cutoff: 15 September 2017.

Table S4: Safety Summary.

Patients — no. (%)	**ABCP (n=393)**	**BCP (n=394)**
Patients with ≥ 1 AE	385 (98.0)	390 (99.0)
Grade 3–4 AEs	242 (61.6)	230 (58.4)
Grade 5 AEs	23 (5.9)	21 (5.3)
Treatment-related AEs*	371 (94.4)	376 (95.4)
Treatment-related Grade 3–4 AEs	219 (55.7)	188 (47.7)
Treatment-related Grade 5 AEs	11 (2.8)	9 (2.3)
Serious AEs	165 (42.0)	134 (34.0)
Treatment-related serious AEs*	100 (25.4)	76 (19.3)
AEs leading to withdrawal from any treatment*	128 (32.6)	98 (24.9)
AEs leading to withdrawal from chemotherapy	64 (16.3)	58 (14.7)

* Incidence of treatment-related AEs, serious treatment-related AEs and AEs leading to withdrawal from any treatment are for any treatment.
ABCP denotes atezolizumab + bevacizumab + carboplatin + paclitaxel; AE, adverse event; BCP, bevacizumab + carboplatin + paclitaxel.
Data cutoff: 15 September 2017.

Table S5. Treatment-Related Adverse Events.*

	ABCP (n=393)			BCP (n=394)		
Patients — no. (%)	**Grade 1–2**	**Grade 3–4**	**Grade 5**	**Grade 1–2**	**Grade 3–4**	**Grade 5**
Treatment-related AEs	141 (35.9)	219 (55.7)	11 (2.8)	179 (45.4)	188 (47.7)	9 (2.3)
Treatment-related AEs with an incidence of ≥ 10% in any arm, grade 3–4 severity with incidence of ≥ 1% in any arm, or grade 5 severity						
Alopecia	183 (46.6)	0	0	173 (43.9)	0	0
Peripheral neuropathy	141 (35.9)	11 (2.8)	0	113 (28.7)	9 (2.3)	0
Nausea	119 (30.3)	15 (3.8)	0	101 (25.6)	8 (2.0)	0
Fatigue	88 (22.4)	13 (3.3)	0	79 (20.1)	10 (2.5)	0
Anemia	70 (17.8)	24 (6.1)	0	71 (18.0)	23 (5.8)	0
Decreased appetite	77 (19.6)	10 (2.5)	0	56 (14.2)	3 (0.8)	0
Diarrhea	70 (17.8)	11 (2.8)	0	58 (14.7)	2 (0.5)	0
Neutropenia	18 (4.6)	54 (13.7)	0	24 (6.1)	44 (11.2)	0
Hypertension	50 (12.7)	25 (6.4)	0	42 (10.7)	25 (6.3)	0
Arthralgia	63 (16.0)	3 (0.8)	0	55 (14.0)	4 (1.0)	0
Constipation	65 (16.5)	0	0	45 (11.4)	0	0
Asthenia	52 (13.2)	5 (1.3)	0	53 (13.5)	11 (2.8)	0
Epistaxis	50 (12.7)	4 (1.0)	0	68 (17.3)	0	0
Vomiting	50 (12.7)	6 (1.5)	0	51 (12.9)	5 (1.3)	0
Decreased platelet count	34 (8.7)	20 (5.1)	0	35 (8.9)	9 (2.3)	0
Myalgia	51 (13.0)	2 (0.5)	0	46 (11.7)	1 (0.3)	0
Thrombocytopenia	36 (9.2)	16 (4.1)	0	28 (7.1)	17 (4.3)	0
Proteinuria	41 (10.4)	10 (2.5)	0	37 (9.4)	11 (2.8)	0
Decreased neutrophil count	14 (3.6)	34 (8.7)	0	10 (2.5)	25 (6.3)	0

Rash	47 (12.0)	5 (1.3)	0	20 (5.1)	0	0
Stomatitis	43 (10.9)	4 (1.0)	0	20 (5.1)	1 (0.3)	0
Paresthesia	42 (10.7)	0	0	36 (9.1)	1 (0.3)	0
Febrile neutropenia	2 (0.5)	33 (8.4)	3 (0.8)	0	23 (5.8)	0
Decreased white blood cell count	14 (3.6)	13 (3.3)	0	10 (2.5)	11 (2.8)	0
Decreased weight	20 (5.1)	4 (1.0)	0	18 (4.6)	1 (0.3)	0
ALT increased	17 (4.3)	4 (1.0)	0	12 (3.0)	1 (0.3)	0
Dehydration	9 (2.3)	8 (2.0)	0	11 (2.8)	4 (1.0)	0
AST increased	16 (4.1)	4 (1.0)	0	8 (2.0)	1 (0.3)	0
Leukopenia	6 (1.5)	7 (1.8)	0	11 (2.8)	4 (1.0)	0
Hemoptysis	15 (3.8)	0	3 (0.8)	8 (2.0)	0	0
Hypokalemia	9 (2.3)	7 (1.8)	0	5 (1.3)	2 (0.5)	0
Pulmonary embolism	2 (0.5)	7 (1.8)	0	2 (0.5)	8 (2.0)	2 (0.5)
Hyponatremia	4 (1.0)	8 (2.0)	0	3 (0.8)	4 (1.0)	0
Pneumonia	3 (0.8)	7 (1.8)	0	2 (0.5)	2 (0.5)	1 (0.3)
Pneumonitis	5 (1.3)	4 (1.0)	0	0	0	0
Colitis	3 (0.8)	5 (1.3)	0	0	0	0
Transaminases increased	3 (0.8)	4 (1.0)	0	1 (0.3)	0	0
Pulmonary hemorrhage	0	0	2 (0.5)	2 (0.5)	0	2 (0.5)
Cerebrovascular accident	1 (0.3)	1 (0.3)	1 (0.3)	0	0	0
Intestinal perforation	0	0	0	0	0	2 (0.5)
Posterior reversible encephalopathy syndrome	0	0	0	0	1 (0.3)	1 (0.3)
Sepsis	0	1 (0.3)	0	0	0	1 (0.3)
Aortic dissection	0	0	1 (0.3)	0	0	0

Intestinal obstruction	0	0	1 (0.3)	0	0	0

* Incidence of treatment-related AEs for any treatment.
ABCP denotes atezolizumab + bevacizumab + carboplatin + paclitaxel; AE, adverse event; ALT, alanine aminotransferase; AST, aspartate aminotransferase; BCP, bevacizumab + carboplatin + paclitaxel.
Data cutoff: 15 September 2017.

Table S6. Serious Treatment-Related Adverse Events.*

	ABCP (n=393)			BCP (n=394)		
Patients — no. (%)	**Grade 1–2**	**Grade 3–4**	**Grade 5**	**Grade 1–2**	**Grade 3–4**	**Grade 5**
Serious treatment-related AEs	15 (3.8)	74 (18.8)	11 (2.8)	9 (2.3)	58 (14.7)	9 (2.3)
Febrile neutropenia	1 (0.3)	21 (5.3)	3 (0.8)	0	15 (3.8)	0
Pneumonia	1 (0.3)	6 (1.5)	0	0	2 (0.5)	1 (0.3)
Anemia	1 (0.3)	4 (1.0)	0	1 (0.3)	3 (0.8)	0
Dehydration	1 (0.3)	5 (1.3)	0	1 (0.3)	2 (0.5)	0
Nausea	1 (0.3)	5 (1.3)	0	0	3 (0.8)	0
Diarrhea	0	5 (1.3)	0	1 (0.3)	2 (0.5)	0
Pulmonary embolism	0	2 (0.5)	0	1 (0.3)	2 (0.5)	2 (0.5)
Thrombocytopenia	1 (0.3)	4 (1.0)	0	0	2 (0.5)	0
Vomiting	2 (0.5)	2 (0.5)	0	0	3 (0.8)	0
Hemoptysis	2 (0.5)	0	3 (0.8)	1 (0.3)	0	0
Neutropenia	0	4 (1.0)	0	0	2 (0.5)	0
Pneumonitis	2 (0.5)	4 (1.0)	0	0	0	0
Pulmonary hemorrhage	0	0	2 (0.5)	1 (0.3)	0	2 (0.5)
Colitis	1 (0.3)	4 (1.0)	0	0	0	0
Pancytopenia	0	2 (0.5)	0	0	3 (0.8)	0
Platelet count decreased	0	2 (0.5)	0	0	2 (0.5)	0
Acute kidney injury	1 (0.3)	1 (0.3)	0	0	1 (0.3)	0
Cerebrovascular accident	1 (0.3)	1 (0.3)	1 (0.3)	0	0	0
Hypersensitivity	0	1 (0.3)	0	1 (0.3)	1 (0.3)	0
Transient ischemic attack	2 (0.5)	0	0	1 (0.3)	0	0

White blood cell count decreased	0	2 (0.5)	0	0	1 (0.3)	0
Abdominal pain	0	1 (0.3)	0	0	1 (0.3)	0
Cardiac failure	0	1 (0.3)	0	0	1 (0.3)	0
Deep vein thrombosis	1 (0.3)	0	0	0	1 (0.3)	0
Diverticulitis	0	1 (0.3)	0	0	1 (0.3)	0
Dyspnoea	1 (0.3)	0	0	0	1 (0.3)	0
General physical health deterioration	0	0	0	1 (0.3)	1 (0.3)	0
Hepatitis	0	2 (0.5)	0	0	0	0
Intestinal perforation	0	0	0	0	0	2 (0.5)
Ischemic colitis	0	0	0	0	2 (0.5)	0
Large intestine perforation	1 (0.3)	1 (0.3)	0	0	0	0
Lung infection	0	1 (0.3)	0	0	1 (0.3)	0
Peripheral neuropathy	0	1 (0.3)	0	1 (0.3)	2 (0.5)	0
Posterior reversible encephalopathy syndrome	0	0	0	0	1 (0.3)	1 (0.3)
Pyrexia	1 (0.3)	1 (0.3)	0	0	0	0
Respiratory tract infection	0	2 (0.5)	0	0	0	0
Sepsis	0	1 (0.3)	0	0	0	1 (0.3)
Syncope	1 (0.3)	0	0	0	1 (0.3)	0
Urinary tract infection	0	1 (0.3)	0	0	1 (0.3)	0
Acute myocardial infarction	0	1 (0.3)	0	0	0	0
Adrenal insufficiency	0	1 (0.3)	0	0	0	0
Anal abscess	0	0	0	0	1 (0.3)	0
Aortic dissection	0	0	1 (0.3)	0	0	0

Arthralgia	0	0	0	0	1 (0.3)	0
Ataxia	0	0	0	0	1 (0.3)	0
Back pain	0	0	0	0	1 (0.3)	0
Blood pressure increased	0	0	0	1 (0.3)	0	0
Bone marrow failure	0	0	0	0	1 (0.3)	0
Cardiac failure congestive	0	0	0	0	1 (0.3)	0
Cardiac ventricular thrombosis	0	1 (0.3)	0	0	0	0
Cerebral hemorrhage	0	0	0	0	1 (0.3)	0
Cerebral ischemia	1 (0.3)	0	0	0	0	0
Chest pain	0	0	0	0	1 (0.3)	0
Cholangitis	0	1 (0.3)	0	0	0	0
Chronic obstructive pulmonary disease	0	1 (0.3)	0	0	0	0
Clostridium difficile infection	0	1 (0.3)	0	0	0	0
Compartment syndrome	0	1 (0.3)	0	0	0	0
Decreased appetite	0	1 (0.3)	0	0	0	0
Dermatitis	0	1 (0.3)	0	0	0	0
Depressed level of consciousness	1 (0.3)	0	0	0	0	0
Dizziness	0	1 (0.3)	0	0	0	0
Drug hypersensitivity	1 (0.3)	0	0	0	0	0
Duodenal ulcer	0	1 (0.3)	0	0	0	0
Ejection fraction decreased	0	1 (0.3)	0	0	0	0
Encephalitis	0	1 (0.3)	0	0	0	0
Epistaxis	0	0	0	1 (0.3)	0	0
Fecaloma	0	0	0	0	1 (0.3)	0

Fall	0	1 (0.3)	0	0	0	0
Fatigue	0	1 (0.3)	0	0	0	0
Gastritis	1 (0.3)	0	0	0	0	0
Headache	0	1 (0.3)	0	0	0	0
Hematoma	0	1 (0.3)	0	0	0	0
Hematuria	0	0	0	0	1 (0.3)	0
Hepatotoxicity	0	1 (0.3)	0	0	0	0
Hiccups	0	0	0	0	1 (0.3)	0
Hyperglycemia	0	0	0	1 (0.3)	0	0
Hypertension	0	0	0	1 (0.3)	0	0
Hypertensive crisis	0	1 (0.3)	0	0	0	0
Hyponatremia	0	0	0	0	1 (0.3)	0
Hypophysitis	0	1 (0.3)	0	0	0	0
Infective exacerbation of chronic obstructive airways disease	0	1 (0.3)	0	0	0	0
Influenza-like illness	0	0	0	1 (0.3)	0	0
Infusion-related reaction	0	1 (0.3)	0	0	0	0
Interstitial lung disease	0	1 (0.3)	0	0	0	0
Intestinal hemorrhage	0	1 (0.3)	0	0	0	0
Intestinal obstruction	0	0	1 (0.3)	0	0	0
Ischemic stroke	0	0	0	1 (0.3)	0	0
Large intestinal hemorrhage	0	0	0	0	1 (0.3)	0
Left ventricular dysfunction	0	0	0	0	1 (0.3)	0
Leukopenia	0	0	0	0	1 (0.3)	0
Lipase increased	0	1 (0.3)	0	0	0	0

Metabolic acidosis	0	1 (0.3)	0	0	0	0
Mucosal inflammation	0	1 (0.3)	0	0	0	0
Muscular weakness	0	1 (0.3)	0	0	0	0
Myalgia	0	1 (0.3)	0	0	0	0
Myocardial infarction	0	1 (0.3)	0	0	0	0
Neutrophil count decreased	0	0	0	0	1 (0.3)	0
Pleurisy	0	1 (0.3)	0	0	0	0
Pyopneumothorax	1 (0.3)	0	0	0	0	0
Rash	0	1 (0.3)	0	0	0	0
Renal failure	0	1 (0.3)	0	0	0	0
Septic shock	0	1 (0.3)	0	0	0	0
Staphylococcal bacteremia	0	1 (0.3)	0	0	0	0
Staphylococcal infection	0	0	0	0	1 (0.3)	0
Tubulointerstitial nephritis	1 (0.3)	0	0	0	0	0
Wound complication	0	0	0	0	1 (0.3)	0

* Incidence of serious treatment-related AEs are for any treatment.
ABCP denotes atezolizumab + bevacizumab + carboplatin + paclitaxel; AE, adverse event; BCP, bevacizumab + carboplatin + paclitaxel.
Data cutoff: 15 September 2017.

Table S7: Immune-Related Adverse Events.*

No. of Patients (%)	**ABCP (N=393)**	**BCP (N=394)**
Rash		
All grades	113 (28.8)	52 (13.2)
Grade 3–4	9 (2.3)	2 (0.5)
Hepatitis (laboratory abnormalities)		
All grades	47 (12.0)	29 (7.4)
Grade 3–4	16 (4.1)	3 (0.8)
Hypothyroidism		
All grades	50 (12.7)	15 (3.8)
Grade 3–4	1 (0.3)	0
Hyperthyroidism		
All grades	16 (4.1)	5 (1.3)
Grade 3–4	1 (0.3)	0
Pneumonitis		
All grades	11 (2.8)	5 (1.3)
Grade 3–4	6 (1.5)	2 (0.5)
Colitis		
All grades	9 (2.3)	2 (0.5)
Grade 3–4	5 (1.3)	2 (0.5)
Hepatitis (diagnosis)		
All grades	8 (2.0)	0
Grade 3–4	4 (1.0)	0
Severe cutaneous reaction		
All grades	4 (1.0)	1 (0.3)
Adrenal insufficiency		
All grades	2 (0.5)	3 (0.8)
Grade 3–4	1 (0.3)	1 (0.3)
Pancreatitis		
All grades	5 (1.3)	0
Grade 3–4	2 (0.5)	0
Hypophysitis		
All grades	3 (0.8)	0
Grade 3–4	1 (0.3)	0
Nephritis		
All grades	3 (0.8)	0
Grade 3–4	1 (0.3)	0
Ocular inflammatory toxicity		
All grades	3 (0.8)	0
Grade 3–4	1 (0.3)	0
Myositis		
All grades	2 (0.5)	1 (0.3)
Grade 3–4	1 (0.3)	0
Autoimmune hemolytic anemia		
All grades	1 (0.3)	1 (0.3)
Vasculitis		
All grades	1 (0.3)	1 (0.3)

Diabetes mellitus		
All grades	1 (0.3)	0
Grade 3–4	0	0
Encephalitis		
All grades	1 (0.3)	0
Grade 3–4	1 (0.3)	0
Meningoencepalitis		
All grades	1 (0.3)	0
Grade 3–4	1 (0.3)	0

* Immune-related AEs were defined using MedDRA Preferred Terms that included both diagnosed immune conditions and signs and symptoms potentially representative of immune-related events, regardless of investigator-assessed causality.
ABCP denotes atezolizumab + bevacizumab + carboplatin + paclitaxel; BCP, bevacizumab + carboplatin + paclitaxel.
Data cutoff: 15 September 2017.